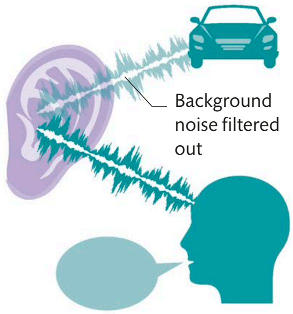
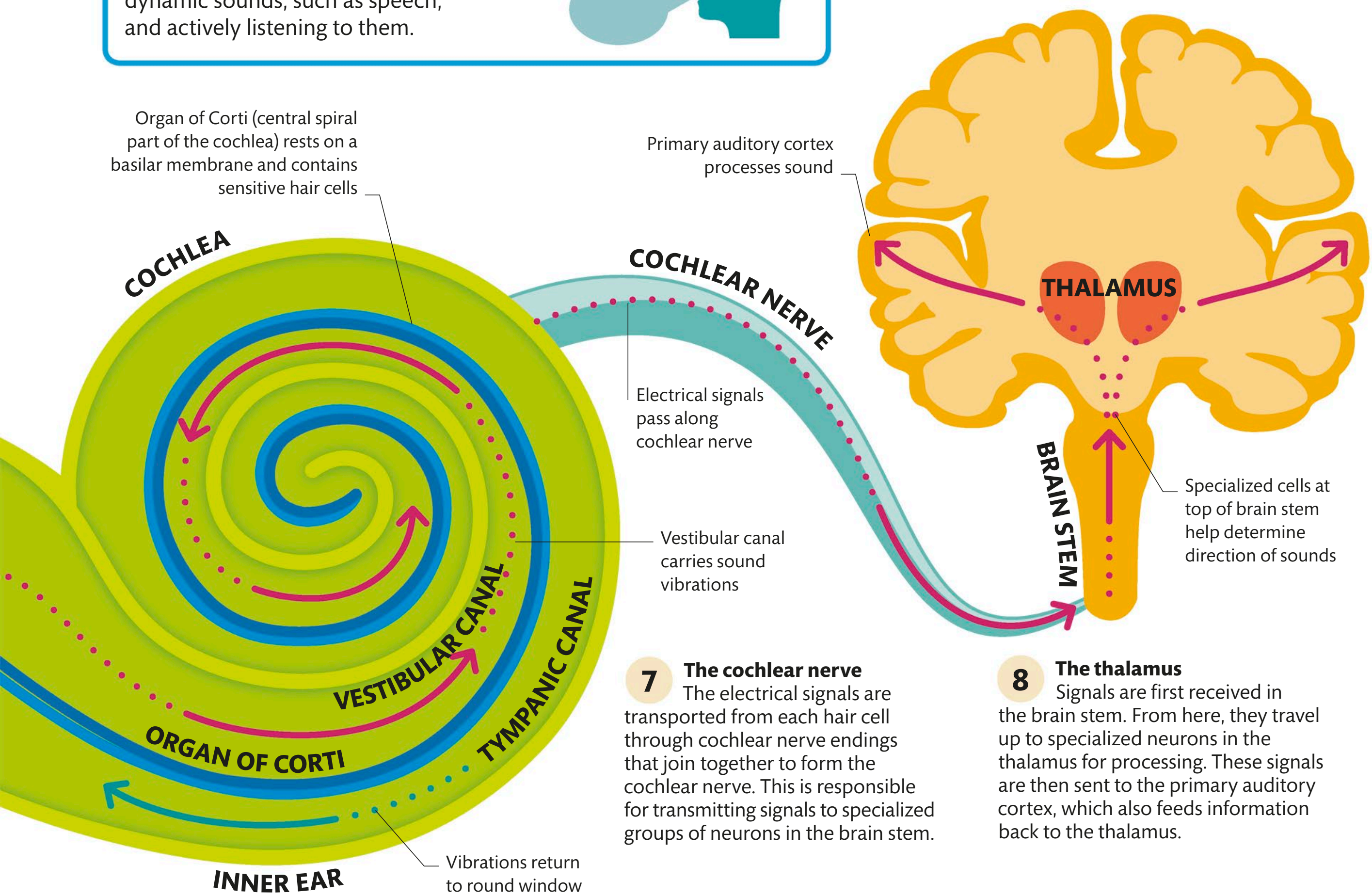
FILTERING OUT NOISE

On a busy street, there are lots of conflicting sounds, yet you can still hear someone talking next to you. This is because the primary auditory cortex can filter out unnecessary sounds and boost the signals it wants to hear. It does this by dampening the response to sustained sounds, such as traffic, while enhancing more dynamic sounds, such as speech, and actively listening to them.





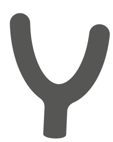
9 The primary auditory cortex
After intermediate processing in the thalamus, the characteristics of each sound are interpreted by the primary auditory cortex, which works with other cortical areas to identify the type of sound.

7 The cochlear nerve
The electrical signals are transported from each hair cell through cochlear nerve endings that join together to form the cochlear nerve. This is responsible for transmitting signals to specialized groups of neurons in the brain stem.

8 The thalamus
Signals are first received in the brain stem. From here, they travel up to specialized neurons in the thalamus for processing. These signals are then sent to the primary auditory cortex, which also feeds information back to the thalamus.

5 The cochlea
The cochlea contains three fluid-filled ducts. Vibrations travel along the vestibular canal as wavelike movements that are transferred to the basilar membrane of the organ of Corti. Residual vibrations return along the tympanic canal to the round window.

6 The organ of Corti
The movement of the basilar membrane bends sensitive hair cells in the organ of Corti (see p.76), which is the main organ of hearing. The hair cells convert this movement into electrical signals.



THE STAPES IS THE SMALLEST BONE IN THE BODY